Predicting Predictability

When can a neural network skip the simulation?

Research Internship proposal · Lossfunk

1. The What

What I am claiming

A neural surrogate for a dynamical system is a shortcut. Instead of running a solver for 64 steps you do one forward pass. Sometimes this works well. Sometimes the model falls apart after two steps and no amount of tuning fixes it. The field treats that second case as an engineering problem: change the architecture, add data, scale up.

I claim it is often not an engineering problem. How far a surrogate can leap ahead before it breaks is largely a property of the *system*, not of the model, and it can be measured before training anything.

The sharper version is that learnability is a separate axis from chaos. The natural assumption is that chaotic systems are hard to predict and stable ones are easy, so the largest Lyapunov exponent should tell you what you need. I think it does not. Two systems can be equally chaotic at the level of individual trajectories while one has a self-contained coarse description a network can latch onto and the other does not. If that holds, chaoticity and learnability come apart, and the second one is what matters for surrogate modelling.

Concretely: sweep a large family of systems, measure one number per system (how many simulator steps a fixed-budget surrogate can jump before error crosses a threshold), and test which measurable properties of the system predict that number.

The surprise

- **Surrogate failure is treated as a modelling failure.** If a system's leap horizon saturates as you scale model size and data, that is an intrinsic ceiling, not underfitting. Nobody currently checks, so failures get blamed on the architecture.
- **The Lyapunov exponent is treated as the relevant quantity.** Weather and climate ML lean on this heavily. If a large share of the variance in learnability survives after regressing it out, that assumption is incomplete.
- **Computational irreducibility is discussed as binary.** Measuring it as a continuous quantity across rule space turns a philosophical claim into an empirical map.

Who updates: people building neural PDE surrogates and neural operators, people building world models for planning, the artificial life and cellular automata community, and the weather and climate ML groups who spend real money deciding which regimes are worth modelling.

One sentence

I want to find out whether how far a neural network can fast-forward a dynamical system is an intrinsic, measurable property of that system rather than a property of the model, and whether it is distinct from how chaotic the system is.

Alignment with Lossfunk

Two of the four directions. **Creativity in artificial systems**, under world models, representations and abstractions: this asks what abstractions a system admits at all, which sits underneath the question of how to build better world models. **Biological foundations of intelligence**, under artificial life and predictive processing: a predictive system is only useful where prediction is possible, and this project draws that boundary.

There is also a direct connection to work already coming out of the lab. *Hybrid Neural World Models* trains one network to leap any number of simulator steps and uses step-doubling disagreement to say when to trust a prediction. My question is the one underneath it: when is the leap possible at all, and can you tell in advance? The trajectory data from that paper is public, so I can validate a predictor on systems the lab already understands.

2. The Why

New questions this opens

- **If closure of a coarse description predicts learnability, can the coarse description be learned instead of hand-picked?** That reframes representation learning from "what latent lowers the loss" to "what latent makes the dynamics closed", which is a testable target.
- **Can compute be allocated by local irreducibility?** If reducibility varies across space inside one system, a surrogate should spend more compute where the dynamics are irreducible. The trust maps in the lab's own work already show error concentrating on shock fronts and wave boundaries.
- **Is there a phase boundary in rule space?** Langton's lambda claimed complexity peaks at the edge of chaos. If learnability drops off a cliff somewhere specific, that is a sharper and measured version of the same claim.
- **Does the same axis predict RL sample efficiency?** If an environment is irreducible, model-based RL should not beat model-free RL there. That is a concrete prediction others can check.

The first two are not askable until a measurable notion of learnability exists. The last two are askable in principle but unanswerable now, because nobody has the map.

Who would build on this

Neural PDE and operator groups, who currently pick benchmarks by convenience rather than by difficulty. The artificial life and NCA community, where irreducibility is discussed constantly and measured almost never. Computational mechanics, where excess entropy and statistical complexity have never been tested against neural learnability. Weather and climate ML, where deciding which regimes and lead times are worth modelling is expensive. And model-based RL researchers, for environment selection and for explaining rollout failure.

Crossroads or corridor

Crossroads. The measurement is shared but four communities want different things from it: physicists want the link to irreducibility, ML people want a pre-training predictor that saves compute, ALife people want the rule-space map, and RL people want to know when a world model is worth building. The failure mode I want to avoid is producing a benchmark, which is a corridor. The map plus a validated predictor is the crossroads version, because the predictor is the part others can run on their own systems.

3. The How

Killer alternative explanation

The obvious dismissal: **your models are just too small, and "irreducibility" is underfitting with a nicer name.**

This is the single most important thing to foreclose and it has a clean test. For a subset of systems spanning the measured range, sweep model capacity across roughly two orders of magnitude and sweep training set size similarly, then look at the leap horizon. If the horizon keeps climbing with capacity, I am measuring my model and the project is dead in that form. If it saturates, and saturates at different levels for different systems, the ceiling belongs to the system. That is the crux experiment and I would run it early rather than late, because it decides whether the rest is worth doing.

Four more, and how each is closed:

- **It is just the Lyapunov exponent.** Treated as the null model. Fit it first, report the variance it explains, claim nothing unless the residual is large and structured. Preregistered kill criterion: if Lyapunov exponent plus entropy rate explains 90% or more of the variance in leap horizon, I report that and stop.
- **It is a cellular automata artifact.** Continuous systems are included from the start, not bolted on later.
- **You picked the rules that worked.** Rule space is sampled on a grid fixed up front, and the whole map is reported including the boring parts.
- **Your error metric hides things.** Pattern-forming systems break MSE badly: a surrogate can produce the right structures at the wrong phase and score terribly while being useful. I report per-cell error, pattern-level statistics and a structure metric, and only claim results that hold across all three.

Experimental design

Measurement. For each system, train a fixed-architecture surrogate to predict the state k steps ahead in one pass, across a horizon ladder. The leap horizon is where normalised error crosses a fixed threshold. That single number per system is the dependent variable.

Substrate 1, rule sweep. Cellular automata and neural cellular automata. I already have a large sequence dataset from earlier work, which gives me a rule library with controllable complexity at very low compute. This is where the map gets built.

Substrate 2, continuous systems. Coupled map lattices with tunable coupling, Lorenz-96 with tunable forcing, Kuramoto-Sivashinsky with tunable domain size, and Gray-Scott reaction diffusion across its regimes. Each has a knob that moves the system through qualitatively different behaviour, so predictors can be tested along a continuum rather than across unrelated systems.

Substrate 3, external validation. Public trajectory data from Hybrid Neural World Models: reaction diffusion, compressible flow, rigid body. Predictors fit on substrates 1 and 2 are tested here without refitting.

Candidate predictors, in increasing order of theoretical interest: largest Lyapunov exponent (the null model); entropy rate and spatial correlation length; excess entropy, meaning mutual information between past and future; and closure of a coarse description, meaning how much extra predictive power the microstate adds over a coarse macrostate. If it adds nothing the coarse law is closed, and I expect closed systems to be the learnable ones.

Success criterion. A predictor fit on one part of rule space that generalises to held-out rule families and to substrate 3, beating the Lyapunov baseline by a margin larger than seed variance.

Compute. Small. Grids at or below 128 by 128, surrogates in the small-CNN range, hundreds of short runs rather than a few long ones. A single GPU is enough.

What rigor looks like here

- Three to five seeds per system, leap horizon reported as a distribution and not a point.
- Predictor grid and metric definitions frozen before the sweep runs, so the map cannot be reverse-engineered into a story.
- Held-out rule families rather than held-out samples, since the claim is about generalisation across systems.
- Partial correlations between predictors, because correlation length and excess entropy will overlap.
- A preregistered kill criterion, stated above, and negative regions of the map reported at the same size as positive ones.

What only I can do

Agents handle the sweeps well. Generating rule families, launching hundreds of short training runs, computing Lyapunov exponents and fitting regressions are all automatable, and I would automate them.

What is not: deciding what counts as a fair coarse-graining, which is the conceptual core and a judgment call rather than a search. Recognising "wrong but right" predictions, where a surrogate produces the correct structures at the wrong phase and MSE calls that the same as noise. And interpreting the residual left after the null model, since working out what that structure is corresponds to the actual finding.

Estimated human time: roughly 250 to 350 hours across six months, concentrated in metric design, failure inspection and interpretation.

4. The So What

Impact type

Conceptual, primarily. It moves computational irreducibility from a claim about rules into a measurable property of systems, and separates learnability from chaoticity. **Practical**, secondarily: if a cheap statistic predicts whether a surrogate will work, you can decide before spending the training budget. That matters in climate, molecular dynamics and plasma modelling, where surrogate projects run for months before anyone finds out they were never going to work. **Methodological** as a side effect, since the leap horizon is a cheap, comparable difficulty measure the surrogate literature currently lacks.

Broadest truthful audience

Anyone who has asked what can be predicted without being simulated: forecasters, physicists working on coarse-graining, and cognitive scientists interested in predictive processing, since a brain doing prediction faces the same constraint. The framing that reaches them without overstating is that some systems can be fast-forwarded and some cannot, and we can now measure which is which instead of finding out the hard way.

The science version

Title: *A measurable limit on what can be predicted without simulating it.*

Press release lead: machine learning models are increasingly used to skip ahead through expensive simulations of weather, chemistry and physics. This work shows that whether the skip is possible is a property of the system being simulated rather than of the model doing the skipping, and that it can be measured cheaply in advance. It also finds that how chaotic a system is tells you less about this than expected.

Six-month plan

Weeks	Work
1 to 3	Pipeline. Horizon-ladder surrogate on one system, sanity checks, metric definitions frozen.
4 to 8	Rule sweep on the CA and NCA substrate. First learnability map.
9 to 12	Capacity and data saturation test. The crux experiment. Go or no-go.
13 to 18	Continuous systems. Predictor fitting, held-out rule families, partial correlations.
19 to 22	External validation on public trajectory data. Failure analysis.
23 to 26	Writing, negative results section, released map and code.